EAT NO. CT- 21090

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR (BACHELOR OF SCIENCE IN Computer Science & Information Technology)

FALL SEMESTER EXAMINATIONS 2023

Batch 2021

Time: 3 Hours Dr Usman Amjad Dated:22-01-2024

Max.Marks:60

Design and Analysis of Algorithms- CT-363

Note:

- Attempt ALL Questions in given order
- You can make any logical assumption and solve the question accordingly

Question No 1: Suppose that in activity selection problem, instead of always selecting the first activity to finish, we instead select the last activity to start that is compatible with all previously selected activities. **Describe** how this approach is a greedy algorithm by using a set of 12 activities, assuming start and finish time. **[CLO-1, 10 Marks]**

Question No 2: Explain what is a convex hull and how it is formed? Illustrate a suitable algorithm for finding out convex hull using a suitable example. **[CLO-1, 10 Marks]**

Question No 3: The GCD, or Greatest Common Divisor, is the largest positive integer that divides each of the given integers without leaving a remainder. GCD (8, 12) is 4 because 4 is the largest positive integer that divides both 8 and 12 evenly (without leaving a remainder). **Apply** a recursive function for calculating GCD of two numbers and find its time complexity using recurrences. Show all necessary steps. **[CLO-2, 10 Marks]**

Question No 4: Consider a modification of the rod-cutting problem in which, in addition to a price pi for each rod, each cut incurs a fixed cost of c. The revenue associated with a solution is now the sum of the prices of the pieces minus the costs of making the cuts. Apply a dynamic-programming algorithm to solve this modified problem. **[CLO-2, 10 Marks]**

Question No 5: Compare different algorithm design strategies studied in this course and suggest which strategy is useful in which sort of problems by using suitable examples. **[CLO-3, 10 Marks]**

Question No 6: Given the following set of numbers: [21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3,7], **solve** the BST using AVL property by inserting each number one by one in the given order. After each insertion, draw the resulting AVL tree and specify the balance factor at each node. Ensure that the tree remains balanced after each insertion. Explain the rotations performed, if any, and justify why they were necessary to maintain the AVL tree property. Finally, provide the final AVL tree after all the numbers have been inserted. **[CLO-3, 10 Marks]**

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THIRD YEAR (COMPUTER SCIENCE & INFORMATION TECHNOLOGY/
COMPUTER SCIENCE SPECIALIZATION IN "CYBER SECURITY")
FALL SEMESTER EXAMINATIONS 2024
Batch - 2022

Time: 3 Hours

Dr. Usman Amjad

Dated:04-12-2024
Max.Marks:60

DESIGN AND ANALYSIS OF ALGORITHMS – CT-363

Note:

- Attempt ALL Questions in given order
- You can make any logical assumption and solve the question accordingly

Question No 1: Explain what is a convex hull and how it is formed? Illustrate a suitable algorithm for finding out convex hull using a suitable example. **[CLO-1, 12 Marks]**

Question No 2: You are climbing a staircase. It takes n steps to reach the top. Each time you can either climb 1 or 2 steps. Given the value of n, write an algorithm to compute distinct ways for climbing to the top? Give worst case complexity for the algorithm you have formulated. Now **apply** some algorithm design strategy so that the complexity may be reduced/improved. (You need to write both the algorithms and their complexities). **[CLO-2, 12 Marks]**

Question No 3: Apply Huffman encoding to generate codes on following data. Generate a string pattern from your CNIC number in such a way that for digits 0, 1 – A; 2, 3 – B; 4,5 – C; 6,7 – D; 8,9 – E. Example if your CNIC number is 42201-1177994-6 than the generated pattern will be **CBBAA AADDEEC D**. (Consider space as a separate character) **[CLO-2, 12 Marks]**

Question No 4: Compare different algorithm design strategies studied in this course and suggest which strategy is useful in which sort of problems by using suitable examples.

[CLO-3, 12 Marks]

Question No 5: Working modulo q = 11, **solve** to find out how many spurious hits does the Rabin-Karp matcher encounter in the text T = 3141592653589793 when looking for the pattern P = 26? Show all necessary steps and the final solution. **[CLO-3, 12 Marks]**

CamScanner

2^n

n^2 Rod

n Fib

SEAT NO. CT-20032

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2020

Time: 3 Hours | **Dated:02-02-2023**
Max.Marks:60

Design and Analysis of Algorithms - CT-363

Note: Attempt all questions.
You can make any logical assumption while solving the questions.

Question No 1: Implement a recursive function for calculating GCD of two numbers and find its time complexity using recurrences. Show all necessary steps. Example GCD (18,12) = 6
[CLO-2, 12 Marks]

Question No 2: Describe an efficient algorithm based on merge sort that will find the element from a set that would be at position k if the elements were sorted. **[CLO-1, 12 Marks]**

Question No 3: Consider a modification of the rod-cutting problem in which, in addition to a price pi for each rod, each cut incurs a fixed cost of c. The revenue associated with a solution is now the sum of the prices of the pieces minus the costs of making the cuts. **Implement** a dynamic-programming algorithm to solve this modified problem. ***[CLO-2, 12 Marks]***

Question No 4: Implement Huffman encoding to generate codes on following data. Generate a string pattern from your CNIC number in such a way that for digits 0, 1 – A; 2,3 – B; 4,5 – C; 6,7 – D; 8,9 – E. Example if your CNIC number is 42201-1188994-6 than the generated pattern will be **CBBAA AADDEEB D**. (Consider space as a separate character) **[CLO-2, 12 Marks]**

CBBAA XAAFE

Question No 5: Analyze which algorithm design strategy do you think will be feasible for these problems? Also suggest a problem which can be solved using same design strategy for each problem case. (You don't need to write the algorithms here) **[CLO-3, 12 Marks]**

a) Consider the problem of making *n* cents change with quarters, dimes, nickels, and pennies, and using the least total number of coins. For example, to make change for 67 cents, we first select a quarter (leaving 42 cents). We next select a second quarter (leaving 17 cents), followed by a dime (leaving 7 cents), followed by a nickel (leaving 2 cents), followed by a penny (leaving 1 cent), followed by a penny.

b) Sorting an array of integers using a comparison-based sorting algorithm such as quicksort.

c) Graph search algorithm that works by building a set of paths from a starting node to all other nodes in the graph, with the shortest path to each node being discovered incrementally. The algorithm maintains a set of tentative distances to each node, with the distance to the starting node being set to 0 and the distance to all other nodes being set to infinity. As the algorithm progresses, it updates the distances to each node by considering all of the edges emanating from the node and selecting the shortest path to the node.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

TSCS/ TSCR, Midterm Examinations Fall 2024

Time: 90 minutes **Design and Analysis of Algorithms (CT-363)** **Max Marks: 20**

Paper A

Note:

Attempt all questions. Show all necessary steps. You can make any logical assumption in solving the questions

Question 1: Explain different cases of algorithm analysis using suitable example? Also discuss average case for Linear Search. **[CLO-1, 5 marks]**

Question 2: Given an array, count how many pairs of elements (i, j) are there such that i < j and arr[i] > arr[j]. For the array {8, 4, 2, 1}, the inversions are (8,4), (8,2), (8,1), (4,2), (4,1), (2,1). Thus, the total number of inversions is 6. **Formulate** an algorithm which takes an array as argument and returns total number of inversions. **[CLO-2, 5 marks]**

Question 3: Solve the equation by recurrence tree method. **[CLO-2, 5 marks]**

$$T(n)= 8T(n/2) + n^2 \; ; \; T(1)=1$$

Question 4: Analyze the following algorithm and give Big-O showing all necessary steps. An example call for the function can be **hanoi(3, 'A', 'C', 'B');** **[CLO-3, 5 marks]**

```
void hanoi(int n, char from_rod, char to_rod, char aux_rod)
{
    if (n == 1) {
      printf("Move disk 1 from rod %c to rod %c\n", from_rod, to_rod);
      return;
    }
    hanoi(n - 1, from_rod, aux_rod, to_rod);
    printf("Move disk %d from rod %c to rod %c\n", n, from_rod, to_rod);
    hanoi(n - 1, aux_rod, to_rod, from_rod);
}
```

TSCS/ TSCR, Midterm Examinations Fall 2024

Time: 90 minutes | **Design and Analysis of Algorithms (CT-363)** | **Max Marks: 20**

Paper B

Note:

Attempt all questions. Show all necessary steps. You can make any logical assumption in solving the questions

Question 1: Explain what are preconditions while doing algorithm analysis. Explain each precondition in detail and where required, give suitable example. **[CLO-1, 5 marks]**

Question 2: Given an array, count how many pairs of elements (i, j) are there such that i < j and arr[i] < arr[j]. For the array {1,2,4,8}, the inversions are (1,2), (1,4), (1,8), (2,4), (2,8), (4,8). Thus, the total number of inversions is 6. **Formulate** an algorithm which takes an array as argument and returns total number of inversions. **[CLO-2, 5 marks]**

Question 3: Solve the equation by recurrence tree method. **[CLO-2, 5 marks]**

$$T(n)= 8T(n/2) + n^2 \; ; \; T(1)=1$$

Question 4: Analyze the following algorithm and give Big-O showing all necessary steps. An example call for the function can be **josephus(n, k);** **[CLO-3, 5 marks]**

```
int josephus(int n, int k) {
  if (n == 1)
    return 0;
  return (josephus(n - 1, k) + k) % n;
}
```